

Clustering UK Universities and Predicting Graduate Outcomes

1 Problem Statement

A dataset containing information on UK universities, sourced from *The Guardian* and the UCAS 2022 report, has been used in the analysis to answer the following research question:

Can universities be grouped based on academic characteristics, and do these groupings explain differences in graduate career outcomes?

An initial investigation has been conducted to determine whether universities can be grouped in clusters. Afterwards, the association between the clusters and post-graduation outcomes has been examined.

2 Methodology

2.1 Data Preprocessing

Only the covariates of interest have been retained from the dataset: *teaching satisfaction*, *feedback satisfaction*, *student-staff ratio*, *spending per student*, *average entry tariff*, *added value*, *continuation* and *career after 15 months*.

Career after 15 months, initially expressed as a percentage, has been transformed into a proportion. In addition, all other variables have been standardised.

The data corresponding to *Cambridge and Falmouth University* have been excluded due to the presence of missing values, leaving $N = 118$ observations in the dataset.

2.2 Principal Component Analysis

Principal Component Analysis (PCA) has been performed on the scaled covariates, not including *career after 15 months*, to reduce dimensionality. The first four principal components (PCs) have been retained, explaining approximately 87% of the total variance.

2.3 Clustering Analysis

Initial Model

A Gaussian Mixture Model (GMM) has been fitted, using JAGS, to the first four PCs in order to evaluate the clustering structure. For each cluster, a different vector of means $\mu^{(k)}$ and a covariance matrix $\Sigma^{(k)}$ have been assumed.

To prevent the number of estimated parameters from becoming too large relative to the number of observations, the covariance matrices have been restricted to diagonal form, implying independence between PCs. This assumption is partially supported by the absence of linear correlations among the principal components.

The model has been fitted using uninformative priors and evaluated under different numbers of clusters, specifically $K = 2, 3, 4$. Model selection has been performed using the DIC, details are provided in Table 1. The best result has been obtained for $K = 2$.

Clustering Problems

An inspection of the clustering results has revealed low effective sample sizes (ESS) for some of the posterior estimates, with values on the order of a few hundred, despite running three chains with 10,000 iterations each, details are provided in Table 2.

This issue is due to the relatively high number of parameters being estimated compared to the limited sample size, specifically: 16 parameters relative to only 118 data points.

A simplified model with fewer parameters is proposed.

Final Clustering Model

Let $\mathbf{Y}_i = (Y_{1i}, Y_{2i})$ and $k \in \{1, 2\}$:

$$Y_i | (Z_i = k) \sim \text{BVN}(\mu^{(k)}, \Sigma^{(k)}) \quad \forall i = 1, \dots, N$$

$$P(Z_i = k) = p_{i,k}, \quad \sum_k p_{i,k} = 1 \quad \forall i = 1, \dots, N$$

with the following priors:

$$\mu^{(k)} \sim \text{BVN}(\mathbf{0}, 5 \cdot \mathbf{I}_2),$$

$$\Sigma^{(k)} = \begin{pmatrix} \sigma_1^{2,(k)} & \sigma_1^{(k)} \sigma_2^{(k)} \rho^{(k)} \\ \sigma_1^{(k)} \sigma_2^{(k)} \rho^{(k)} & \sigma_2^{2,(k)} \end{pmatrix},$$

$$\rho^{(k)} \sim \text{Unif}(-1, 1), \quad \sigma_1^{2,(k)}, \sigma_2^{2,(k)} \sim \text{Inv-Gamma}(1, 1)$$

$$p_i \sim \text{Dirichlet}(1, 1) \quad \text{where} \quad \mathbf{p}_i = (p_{i,1}, p_{i,2})$$

where:

- Z_i is the latent cluster indicator
- All parameters are assumed to be independent a priori

The new model considers only the first two PCs, which capture approximately 60% of the total variance. The assumption of independence between components is relaxed, and the two variables are modelled jointly using a bivariate Normal distribution.

Confidence in the use of PC1 and PC2 as the basis for clustering is supported by the fact that in the previous model, the credible intervals for the clusters' means largely overlapped across all components except for PC1, indicating that most components did not contribute to the clustering structure.

ESS and DIC appear to be significantly improved. Gelman-Rubin statistics show no problem in convergence.

Model	Mean Deviance	Penalty	DIC
$K = 2$	1398.0	114.0	1512.0
$K = 3$	1344.0	199.6	1544.0
$K = 4$	1302.0	249.1	1551.0
Final model	722.2	110.2	832.3

Table 1: DIC values for different models

Parameter	ESS K=2	ESS Final Model
$\mu_1^{(1)}$	666.08	2554.69
$\mu_1^{(2)}$	1495.68	3833.84
$\mu_2^{(1)}$	3174.96	5245.38
$\mu_2^{(2)}$	2862.66	6059.27
$\sigma_1^{(1)}$	490.45	3283.26
$\sigma_1^{(2)}$	3224.07	3792.71
$\sigma_2^{(1)}$	2367.71	5605.60
$\sigma_2^{(2)}$	1371.59	7747.32

Table 2: ESS for K=2 Model and Final Model

2.4 Bayesian Beta Regression

The goal of this model is to estimate the distribution of *career after 15 months* across the two clusters. This is done by modelling the response variable as Beta distributed with distinct mean and variance for each cluster. The model has been fitted in STAN.

Model

Let $c_i \in \{1, 2\}$ be the cluster assignment for the i -th observation, and $y_i \in (0, 1)$ the career after 15 months for observation i .

- $\mu_{c_i} \in (0, 1)$: mean outcome for cluster $c_i \in \{1, 2\}$
- $\sigma_{c_i}^2$: variance outcome for cluster c_i
- ϕ_{c_i} : precision parameter for cluster c_i :

$$\phi_{c_i} = \frac{\mu_{c_i}(1 - \mu_{c_i})}{\sigma_{c_i}^2} - 1$$

- Career after 15 months is modelled as:

$$y_i \sim \text{Beta}(\mu_{c_i} \cdot \phi_{c_i}, (1 - \mu_{c_i}) \cdot \phi_{c_i})$$

Uninformative independent priors have been assigned to the model parameters, for both clusters:

$$\mu_{c_i} \sim \text{Beta}(2, 2), \quad \sigma_{c_i}^2 \sim \text{Inv-Gamma}(0.001, 0.001).$$

The posterior results show excellent mixing.

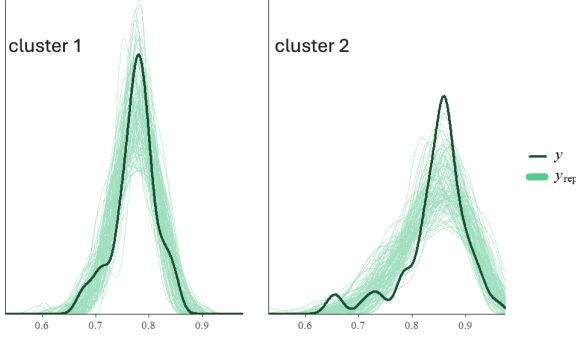


Figure 1: Posterior predictive checks by cluster

Posterior predictive checks have been performed by simulating *career after 15 months* data from the cluster-specific posterior beta distributions and comparing to the observed densities in each cluster. Figure 1 shows the simulated data aligns well with the observed data for both clusters, indicating good model fit. The match is stronger in cluster 1, while cluster 2 shows slightly higher variability.

3 Results and Interpretation

3.1 Clustering

Figure 2 shows the modal cluster assignment for each observation, from the plot, it is clear that the clustering is driven by PC1: a boundary among the clusters is fixed around 0, with higher values associated with cluster 2 and lower values with cluster 1.

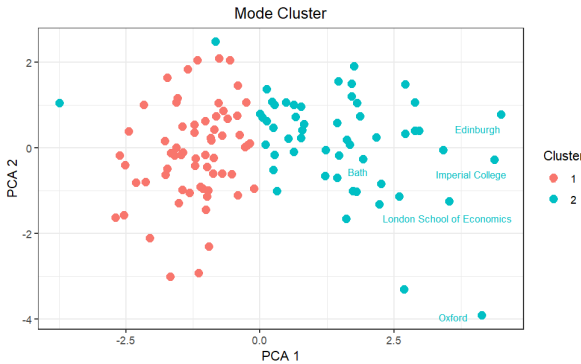


Figure 2: Clustering mode results on PC1 and PC2

The posterior means of the cluster along the PC1 are -1.08 for cluster 1 and 1.06 for cluster 2, with credible intervals that do not overlap, indicating clear separation along this axis. Along PC2, the means show overlapping credible intervals, indicating that the clustering does not strongly depend on it. At the same time, the credible interval for the mean of PC2 for cluster 2 has a higher upper bound that could explain why a point with PC1 below 0 but with the highest value for PC2 is classified as cluster 2.

Regarding PC1, the posterior standard deviations reveal that cluster 1 is more compact, $\sigma_1^{(1)} = 0.84$, while cluster 2 exhibits greater dispersion, $\sigma_1^{(2)} = 1.65$, with credible intervals that do not overlap. This increased variability in the second cluster could be the reason why the observation with the lowest PC1 in the dataset is in cluster 2.

The reliability of the clustering assignments is assessed by evaluating the stability of the cluster labels for each university. For each observation, the proportion of iterations in which the university is assigned to its modal cluster is computed. The quantiles of this measure are then taken into account.

	Min.	1 st Qu.	Median	3 rd Qu.	Max.
Stability	0.5093	0.7109	0.8688	0.9577	1.0000

Table 3: Summary statistics of cluster assignment stability

As shown in Table 3, 25% of the universities are assigned to their modal cluster in fewer than 71% of the samples, while for half of them this proportion is at least 87%.

3.2 Beta Regression

The results of the Beta regression indicate that the posterior distributions of *career after 15 months* differ clearly between the two clusters, as evidenced by non-overlapping posterior credible intervals for both the means and the variances.

The posterior mean and variance are 0.77 and 0.0016 for cluster 1, while for cluster 2 they are 0.84 and 0.0036, indicating that universities with higher values of PC1 tend to exhibit better post-graduation outcomes, but with higher variability.

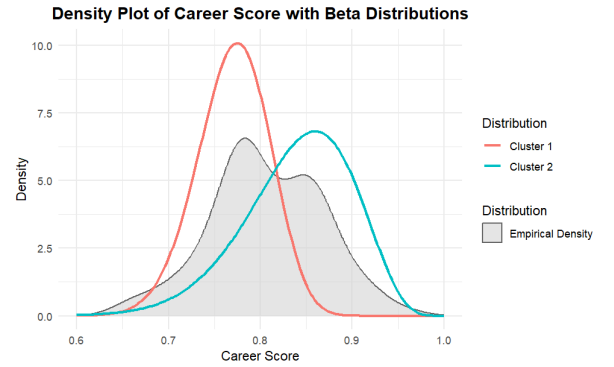


Figure 3: Beta regression density results

The posterior means of the two clusters align with the bimodal structure observed in the empirical distribution of *career after 15 months*.

A simple Monte Carlo simulation shows that the probability of a university in cluster 1 outperforming one in cluster 2 is approximately 16.4%, reinforcing that cluster 2 is associated with better graduate outcomes.

3.3 PCA

The interpretation focuses on PC1, which explains most of the clustering structure. The weights of the original academic covariates for PC1 show large positive values for *continuation*, *average entry tariff*, and *spending per student*, as well as large negative loadings for *student-to-staff ratio* and *feedback satisfaction*. The magnitudes of these values are relatively similar, ranging from 0.37 to 0.5.

These patterns support the interpretation of PC1 as a dimension where higher scores correspond to universities that are better resourced and more selective. Importantly, universities with higher PC1 values tend to exhibit better post-graduation outcomes, as shown by the beta regression results.

The strong negative weight of *feedback satisfaction* (-0.47) appears interesting, as it suggests that universities where students report higher satisfaction with feedback tend to show lower post-graduation outcomes.

4 Conclusion

This study has explored the relationship between academic characteristics and graduate outcomes. It has been found that universities can be grouped into two clusters based on the first principal component, which reflects institutional selectivity and resources. These clusters show different post-graduation outcomes, with better resourced and more selective institutions tending to achieve higher, but more variable, success rates. Academic metrics play a meaningful role in shaping future careers.

Appendix: code for analysis

```
#import Data
data <- read.csv("finalbis.csv",
                 header = TRUE, sep = ",")

data$continuation <- as.numeric(data$continuation)

#only accademic covariate
pca_data <- na.omit(data[,c(4:12)]) #cambridge and Falmouth omitted here

#career_after_15_month as a proportion
pca_data$career_after_15_month <- pca_data$career_after_15_month / 100

#scale covariates, not career_after_15_months or university names
pca_data_scaled <- scale(pca_data[,c(2:7,9)])

#perform PCA
pca <- prcomp(pca_data_scaled)

summary(pca) #first four explain 87% variance

pca$rotation #loadings, for interpretation later

pcs <- pca$x[, 1:4] #use first 4 PCs for clustering

#for later plotting select some well-known high performing unis
selected_universities <- c('Imperial College ', 'Oxford ',
                          'London School of Economics ', 'Bath ', 'Edinburgh ')
university <- pca_data[,1]

selected_labels <- ifelse(university %in% selected_universities,
                          university, "")

pca_data$labels <- selected_labels

#check pca components and variance to get a sense if normal
#distribution is appropriate and what priors for the parameters
hist(pcs[,1])
hist(pcs[,2])
hist(pcs[,3])
hist(pcs[,4])

var(pcs[,1])
var(pcs[,2])
var(pcs[,3])
```

```

var(pcs[,4])

#first model, GMM, K = number of clusters, using first 4 PCs,
#each cluster has its own covariance matrix( diagonal) and mu vectors.
#All parameters are assumed independent a priori.
model_string_clust_cov <- "
model{

  #Priors for cluster means (4 dimensions)
  for(i in 1:k){
    mu1_start[i] ~ dnorm(mu00, tau00)
  }

  #sort mu1 to avoid label switching
  mu1 <- sort(mu1_start)

  mu[1:k, 1] <- mu1

  for(i in 1:k){
    for(j in 2:4){
      mu[i, j] ~ dnorm(mu00, tau00)
    }
  }

  #precision (tau) for each cluster
  #(i in 1:k) and for each PCs (1 : 4)
  for(i in 1:k){
    for(j in 1:4){
      tau[i, j] ~ dgamma(1, 1)
    }

    #diagonal precision matrix for cluster i
    for(j in 1:4){
      for(l in 1:4){
        prec[i, j, l] <- equals(j, l) * tau[i, j]
      }
    }
  }

  # Cluster assignment and likelihood
  for(i in 1:n){

    p[i, 1:k] ~ ddirch(rep(1, k))

    Z[i] ~ dcat(p[i, 1:k])

    #use mean vector and precision
    #matrix for cluster assigned to obs i
    Y[i, 1:4] ~ dmnorm(mu[Z[i], 1:4], prec[Z[i], , ])
  }

```

```

#standard deviations for each cluster and dimension
for(i in 1:k){
  for(j in 1:4){
    sigma[i, j] <- sqrt(1/tau[i, j])
  }
}
}
"

library(rjags)
library(coda)

#function to run the model for different values
#of K and compute DIC

fit_jags_model <- function(k_val, data_list, model_string) {
  cat("Fitting model with k =", k_val, "\n")

  data_list$k <- k_val # numbers of clusters

  jags_model <- jags.model(textConnection(model_string),
                           data = data_list, n.chains = 3)

  update(jags_model, 5000) #burn-in

  # DIC
  dic_val <- dic.samples(jags_model, n.iter = 5000)

  cat("DIC for k =", k_val, "\n")
  print(dic_val)

  return(list(model = jags_model))
}

#run the model with different values of K,
#note this code takes really long time to run

n <- nrow(pcs)

jags_data <- list(
  Y = pcs[, 1:4], #first 4 pca components
  n = n,
  mu00 = 0, tau00 = 1/5 #priors for mu
)

#test with k = 2, 3, 4
results <- list()
ks <- c(2,3,4)

for (k in ks) {
  results[[as.character(k)]] <- fit_jags_model(k, jags_data,
                                              model_string_clust_cov)
}

```

```

}

#dic results:
#k=2 -> mean deviance: 1398, penalty: 114 -> tot:1512
#k=3 -> mean deviance: 1344, penalty: 199.6 -> tot:1544
#k=4 -> mean deviance: 1302, penalty: 249.1 -> tot:1551

#best result is k=2, then we run this model

data_list2<- list(
  Y = pcs[, 1:4], #4 pca components
  n = n,
  mu00 = 0, tau00 = 1/5,
  k=2
)

jags_model_2 <- jags.model(textConnection(model_string_clust_cov),
  data = data_list2,
  n.chains = 3)

update(jags_model_2, 5000) #burn-in

#samples for k=2
samples <- coda.samples(jags_model_2, c("Z","p","mu","sigma"), n.iter=10000)

#summary for each variable
summary_Z <- summary(samples[, grep("^Z", varnames(samples))])
summary_mu <- summary(samples[, grep("^mu", varnames(samples))])
summary_sigma <- summary(samples[, grep("^sigma", varnames(samples))])
summary_p <- summary(samples[, grep("^p", varnames(samples))])

cat("\nSummary per mu:\n")
print(summary_mu)

cat("\nSummary per sigma:\n")
print(summary_sigma)

#Gelman-Rubin per mu
gelman_mu <- gelman.diag(samples[, grep("^mu", varnames(samples))])

#Gelman-Rubin for sigma
gelman_sigma <- gelman.diag(samples[, grep("^sigma", varnames(samples))])

cat("\nGelman-Rubin Diagnostic for mu:\n")
print(gelman_mu)

cat("\nGelman-Rubin Diagnostic for sigma:\n")
print(gelman_sigma)

#ESS mu
ess_mu <- effectiveSize(samples[, grep("^mu", varnames(samples))])

#ESS sigma

```

```

ess_sigma <- effectiveSize(samples[, grep("^sigma", varnames(samples))])

cat("\nEffective Sample Size (ESS) for mu:\n")
print(ess_mu)

cat("\nEffective Sample Size (ESS) for sigma:\n")
print(ess_sigma)

plot(samples[[1]][, "mu[1,1]"]) #slow mixing

#not convinced of this results, fit a new model.
#Use only PC1 and PC2, always GMM,
#each cluster has covariance matrix but now not diagonal.

model_pca12 <- " model{

  #Priors for cluster means

  #PC1
  for(i in 1:k){
    mu1_start[i] ~ dnorm(mu00, tau00)
  }

  #sort to avoid label switching
  mu1 <- sort(mu1_start)
  mu[1:k, 1] <- mu1

  #PC2
  for(i in 1:k){
    mu[i, 2] ~ dnorm(mu00, tau00)
  }

  #precision matrix for each cluster
  for(i in 1:k){

    # precision parameters
    tau[i,1] ~ dgamma(1,1)
    tau[i,2] ~ dgamma(1,1)

    #prior for correlation
    rho[i] ~ dunif(-1,1)

    #precision matrix (inverse of covariance matrix)

    prec[i,1,1] <- tau[i,1] / (1 - pow(rho[i], 2))
    prec[i,2,2] <- tau[i,2] / (1 - pow(rho[i], 2))
    prec[i,1,2] <- -rho[i] * sqrt(tau[i,1] * tau[i,2]) / (1 - pow(rho[i], 2))
    prec[i,2,1] <- prec[i,1,2]
  }

  #cluster assignment and likelihood
  for(i in 1:n){

```

```

#prior for the probs
p[i, 1:k] ~ ddirch(rep(1, k))

#cluster assignment
Z[i] ~ dcat(p[i, 1:k])

#likelihood
Y[i, 1:2] ~ dmnorm(mu[Z[i], 1:2], prec[Z[i], , ])
}

#standard deviations for each cluster and dimension
for(i in 1:k){
  for(j in 1:2){
    sigma[i, j] <- sqrt(1/tau[i, j])
  }
}
}
"

#fit new model

data_list_pca12<- list(
  Y = pcs[, 1:2], #2 pca components
  n = n,
  mu00 = 0, tau00 = 1/5,
  k=2
)

jags_model_pca12 <- jags.model(textConnection(model_pca12),
                              data = data_list_pca12,
                              n.chains = 3)

update(jags_model_pca12, 5000) #burn-in

#samples
samples_pca12 <- coda.samples(jags_model_pca12,
                              c("Z","p","mu","sigma","rho"), n.iter=10000)

summary(samples_pca12[, grep("^mu", varnames(samples_pca12))])
summary(samples_pca12[, grep("^sigma", varnames(samples_pca12))])
summary(samples_pca12[, grep("^rho", varnames(samples_pca12))])

#dic new model
dic_pca12 <- dic.samples(jags_model_pca12, n.iter = 5000)

dic_pca12 #mean deviance 722.2, penalty 110.2, total 832.3

#ESS new model
ess_mu_pca12<- effectiveSize(
  samples_pca12[, grep("^mu", varnames(samples_pca12))])
ess_sigma_pca12 <- effectiveSize(
  samples_pca12[, grep("^sigma", varnames(samples_pca12))])

```



```

ess_rho_pca12<- effectiveSize(
  samples_pca12[, grep("^rho", varnames(samples_pca12))])

ess_mu_pca12
ess_sigma_pca12
ess_rho_pca12

#better mixing? yes
plot(samples_pca12[[1]][, "mu[1,1]"])

#Evaluate cluster assignment stability
#(both old model (K=2) and new model (pca12))

#extract Z (latent clusters labels)

Z_list_old <- lapply(samples, function(chain) {
  chain[, grep("^Z", colnames(chain))]
})

Z_list_pca12 <- lapply(samples_pca12, function(chain) {
  chain[, grep("^Z", colnames(chain))]
})

#put everything in one matrix
Z_samples_old <- do.call(rbind, Z_list_old)

Z_samples_pca12 <- do.call(rbind, Z_list_pca12)

#count proportion of how many times each observation
#is assigned to its modal cluster

cluster_stability_old <- apply(Z_samples_old, 2, function(x) {
  mode_val <- as.numeric(names(which.max(table(x))))
  mean(x == mode_val)
})

cluster_stability_pca12 <- apply(Z_samples_pca12, 2, function(x) {
  mode_val <- as.numeric(names(which.max(table(x))))
  mean(x == mode_val)
})

#print results
summary(cluster_stability_old)
summary(cluster_stability_pca12)

par(mfrow=c(2,2))

#plot results

hist(cluster_stability_old, main = "Cluster Assignment Stability Old Model",
  xlab = "Proportion assigned to modal cluster", breaks = 20,ylim=c(0,35))

```

```

hist(cluster_stability_pca12, main = "Cluster Assignment Stability New Model (pca12)",
      xlab = "Proportion assigned to modal cluster", breaks = 20,ylim=c(0,35))

#compute modal cluster assignment and
#plot results of modal cluster assignments

mode_of_vector <- function(x) {
  frequency <- table(x) # Count occurrences
  mode_value <- as.numeric(names(frequency)[which.max(frequency)])
  return(mode_value)
}

#compute posterior mode for each observation
posterior_mode_Z <- apply(Z_samples_pca12, 2, function(x) mode_of_vector(x))

mode_1_index <- which(posterior_mode_Z == 1) #index of observations
mode_2_index <- which(posterior_mode_Z == 2) #cluster 2

#color vector for plotting
col_tot <- rep(0, n) #set all to 0 initially

#assign colors based on cluster assignment
col_tot[mode_1_index] <- rep(1, length(mode_1_index))
col_tot[mode_2_index] <- rep(2, length(mode_2_index))

library(ggplot2)
library(ggrepel)

#data frame for plotting
plot_data <- data.frame(
  PCA1 = pcs[,1],
  PCA2 = pcs[,2],
  label = pca_data$labels,
  cluster = factor(col_tot)
)

ggplot(plot_data, aes(x = PCA1, y = PCA2, color = cluster)) +
  geom_point(size = 3) + #point coloured for clusters
  geom_text_repel(
    aes(label = label),
    size = 3,           #text size
    box.padding = 0.5,
    max.overlaps = 18, #space around label
    point.padding = 0.5 #space around point
  ) +
  labs(
    title = "Mode Cluster",
    x = "PCA 1",
    y = "PCA 2",

```

```

    color = "Cluster"          # legend title
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5)
  )

#plotting probability of being in cluster 2 vs PC1

# sample matrix
samples_mat <- as.matrix(samples_pca12)

#Extract the column names corresponding to the probability of being in cluster 2
p_cols <- grep("^p\\[\\.*,2\\]", colnames(samples_mat), value = TRUE)

#observation index from the column names
obs_ids <- as.numeric(gsub("p\\[|,2\\]", "", p_cols))

#the mean probability of being in cluster 2 for each observation
p_cluster2_means <- colMeans(samples_mat[, p_cols])

#dataframe with index and their corresponding mean probabilities
prob_cluster2 <- data.frame(
  Observation = obs_ids,
  Mean_Prob_Cluster2 = p_cluster2_means
)

#order by index to get alignment with PC1
prob_cluster2 <- prob_cluster2[order(prob_cluster2$Observation), ]

#order pcs at the same way
pcs_ordered <- pcs[order(prob_cluster2$Observation), ]

#add cluster assignment (color)
prob_cluster2$Group <- factor(col_tot)

#add PC1 to the dataframe
prob_cluster2$PC1 <- pcs_ordered[, 1]

#plot the mean probability of being in cluster 2 against PC1
ggplot(data = prob_cluster2, aes(x = PC1,
                                y = Mean_Prob_Cluster2, color = Group)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "loess", se = FALSE, color = "darkred") + #smoothed curve
  scale_color_manual(values = c("1" = "orangered", "2" = "turquoise")) +
  labs(
    x = "First Principal Component (PC1)",
    y = "Mean Probability of Cluster 2",
    title = "Cluster 2 Probability vs PC1",
    color = "Cluster"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

```

```

#Beta regression to predict career_after15months
#based on the cluster assignment.

model_string1 <-"
data {
  int<lower=1> N;                //number of observations
  int<lower=1, upper=2> cluster[N]; //cluster variable
  vector<lower=0, upper=1>[N] y; //response (career)
}

parameters {
  real<lower=0, upper=1> mu[2]; //mean for each cluster
  real<lower=0> sigma2[2];      //variance for each cluster
}

transformed parameters {
  real<lower=0> phi[2]; //using phi (precision of beta)
  for (c in 1:2) {
    phi[c] = (mu[c] * (1 - mu[c])) / sigma2[c] - 1; //get phi from mu & sigma2
                                                    //(phi = alpha+beta)
  }
}

model {
  //uninformative priors mean and variance in each cluster (same prior)
  mu ~ beta(2, 2);
  sigma2 ~ inv_gamma(0.001, 0.001);

  // likelihood, parametrisation in stan is alpha
  //beta (computed alpha and beta in
  //function of phi (sigma2) and mu )

  for (i in 1:N) {
    y[i] ~ beta(mu[cluster[i]] * phi[cluster[i]],
                (1 - mu[cluster[i]]) * phi[cluster[i]]);
  }
}

"
library(rstan)

fname <- "beta2.stan"
cat(model_string1, file = fname, append = FALSE)

#initail values for the chians
init_fun <- function(chain_id) {
  list(
    mu = c(runif(1, 0.2, 0.4), runif(1, 0.6, 0.8)),
    sigma2 = c(runif(1, 0.01, 0.03), runif(1, 0.03, 0.05))
  )
}

```

```

data_list <- list(
  N = nrow(pca_data),
  cluster = as.numeric(factor(posterior_mode_Z)), #cluster in numbers
  y = pca_data$career_after_15_month # response
)

# fit model
res1_beta <- stan(file = fname, data = data_list,
  iter = 2000, chains = 4, init = init_fun,
  cores = parallel::detectCores())

# results
print(res1_beta, pars = c("mu", "sigma2", "lp_")) #convergence good
#CI of mu do not overlap, we can statistically say
#that clusters are influent on posterior means careers

#cluster 1 shows better perfomance in terms of careers after 15 months

# Traceplots
plot(res1_beta, pars=c("mu"), show_density =TRUE, ci_level = 0.95)
plot(res1_beta, pars=c("sigma2"), show_density =TRUE, ci_level = 0.95)

print(res1_beta, pars = c("sigma2"), digits = 6) # Show 6 decimal places
#CI of sigma2 overlap

plot(res1_beta, par= c("mu", "sigma2"), plotfun="trace")

#posterior predective checks beta regression

library(bayesplot)

#samples
posterior_samples <- rstan::extract(res1_beta)
N <- data_list$N
cluster_assignments <- data_list$cluster

# generate y_rep
y_rep <- matrix(NA, nrow = length(posterior_samples$mu[,1]), ncol = N)
for (s in 1:nrow(y_rep)) {
  for (i in 1:N) {
    c <- cluster_assignments[i]
    alpha <- posterior_samples$mu[s, c] * posterior_samples$phi[s, c]
    beta <- (1 - posterior_samples$mu[s, c]) * posterior_samples$phi[s, c]
    y_rep[s, i] <- rbeta(1, alpha, beta)
  }
}

#plots
ppc_dens_overlay(data_list$y, y_rep[1:100, ]) #density
ppc_stat(data_list$y, y_rep, stat = "mean") #mean
ppc_stat(data_list$y, y_rep, stat = "sd") #sigma

#density per cluster

```

```

ppc_dens_overlay_grouped(data_list$y, y_rep[1:100, ],
                        group = factor(data_list$cluster))

#plot posterior beta distributions

library(dplyr)

#compute Beta parameters from mean and variance
compute_beta_params <- function(mu, var) {
  alpha <- ((mu^2 * (1 - mu)) / var) - mu
  beta <- ((mu * (1 - mu)^2) / var) - (1 - mu)
  return(c(alpha, beta))
}

#from mean and variance for the two clusters get alpha and beta
mu1 <- 0.77
var1 <- 0.001564
alpha1 <- compute_beta_params(mu1, var1)[1]
beta1 <- compute_beta_params(mu1, var1)[2]

mu2 <- 0.84
var2 <- 0.003587
alpha2 <- compute_beta_params(mu2, var2)[1]
beta2 <- compute_beta_params(mu2, var2)[2]

#x values for Beta densities (from 0.6 that is below lower value for career in dataset)
x_vals <- seq(0.6, 1, length.out = 200)

#compute the Beta densities
beta_density1 <- dbeta(x_vals, alpha1, beta1)
beta_density2 <- dbeta(x_vals, alpha2, beta2)

#dataframe for ggplot
beta_data <- data.frame(x = rep(x_vals, 2),
                        density = c(beta_density1, beta_density2),
                        group = rep(c("Beta (mean=0.77)", "Beta (mean=0.85)"),
                                   each = length(x_vals)))

#pca_data to include the empirical density label
pca_data <- pca_data %>%
  mutate(group = "Empirical Density")

#plot the empirical density and Beta densities
ggplot() +
  #empirical density
  geom_density(data = pca_data, aes(x = career_after_15_month, fill = group),
              color = "gray40", alpha = 0.6) +

  #beta densities
  geom_line(data = beta_data, aes(x = x, y = density, color = group), size = 1) +

  #title and labels

```

```

labs(title = "Density Plot of Career Score with Beta Distributions",
     x = "Career Score",
     y = "Density",
     color = "Distribution",
     fill = "Distribution") +

#legend
scale_fill_manual(values = c("Empirical Density" = "lightgray")) +
scale_color_discrete(
labels = c("Cluster 1", "Cluster 2"))+
theme_minimal() +
theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 14)) # Center title

#COMPUTE PROBABILITY THAT CLUSTER 1 UNIVERSITIES HAVE BETTER OUTCOME THEN CLUSTER 2

#mean and variance for the first Beta distribution
mu1 <- 0.77
var1 <- 0.0016

#alpha and beta parameters
alpha1 <- mu1 * ((mu1 * (1 - mu1)) / var1 - 1)
beta1 <- (1 - mu1) * ((mu1 * (1 - mu1)) / var1 - 1)

#mean and variance for the second Beta distribution
mu2 <- 0.84
var2 <- 0.0036

#alpha and beta parameters
alpha2 <- mu2 * ((mu2 * (1 - mu2)) / var2 - 1)
beta2 <- (1 - mu2) * ((mu2 * (1 - mu2)) / var2 - 1)

# Set seed for reproducibility
set.seed(123)

# Number of simulations
N <- 1e6

# Generate random samples from both Beta distributions
x <- rbeta(N, alpha1, beta1)
y <- rbeta(N, alpha2, beta2)

# Estimate the probability that X > Y
prob <- mean(x > y)

#results
print(prob)

```